Experiment – 0

Web Application Design and Planning

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Aim

To analyze a real-world problem, select a suitable web application topic, identify its functional requirements, and design the layout and views of the application before implementation.

Problem Statement

Select any **one** web application topic of your choice. Analyze the system, identify the features and users to be implemented, and prepare the UI layout/ description of all the views/pages that will be developed.

Instructions

1. Select **one** web application topic.
2. Identify the target users.
3. Identify the modules/features of the application.
4. Design the layout of each page/view.
5. Include in short the purpose of every view.
6. Draw wireframes (hand-drawn or using any design tool) for all major views./ Describe the views

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Documentation Format

1. Project Title

Online Auction System

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2. Problem Description

Students at VESIT frequently use WhatsApp groups to sell their second-hand furniture, books, electronics, and other unwanted items. However, these listings often get buried among regular messages, making it difficult for potential buyers to discover relevant products. This results in reduced visibility for sellers and an inefficient buying experience for interested students.

To address this issue, we propose developing a web-based auction platform exclusively for VESIT students. The platform will allow users to upload product images, provide detailed descriptions, set a starting bid price, and list items for auction. Other students can browse available products, place bids, and compete to purchase items in a transparent and organized manner. This system will streamline the buying and selling process while creating a dedicated marketplace for the college community.

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3. Target Users

Roles to be Implemented:

1. **Student (Buyer/Seller)**
    - Register/Login using college credentials.
    - List products for auction.
    - Browse and search available products.
    - Place bids on products.
    - View auction status and bid history.
    - Manage their own listings.
2. **Administrator**
    - Verify and manage user accounts.
    - Approve or remove inappropriate product listings.
    - Monitor ongoing auctions.
    - Resolve disputes and manage reported listings.
    - View platform statistics and activity.

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4. Functional Requirements

Objective:
To define the core system behaviors, features, and user interactions necessary to operate a web-based Online Auction System exclusively for VESIT students. This section outlines the technical capabilities required to transition the student community's second-hand marketplace from unstructured WhatsApp groups to an organized, transparent platform. It details the essential processes for both student and administrator roles, including user management, product listing, active auction management, bidding mechanisms, and automated notifications to streamline the buying and selling process.

Module 1: User Management

- User Registration
- User Login/Logout
- Profile Management
- Password Reset

Module 2: Product Management

- Add Product
- Upload Product Images
- Edit/Delete Product Listing
- View Product Details

Module 3: Auction Management

- Create Auction
- Set Starting Price
- Set Auction Duration
- Automatically Close Auction
- Declare Highest Bidd

Module 4: Bidding System

- Place Bid
- Validate Bid Amount
- Display Current Highest Bid
- View Bid History

Module 5: Search & Browse

- Browse Available Auctions
- Search Products
- Filter by Category or Price

Module 6: Notifications

- Notify User When Outbid
- Notify Auction Winner
- Notify Seller When Auction Ends

Module 7: Admin Module

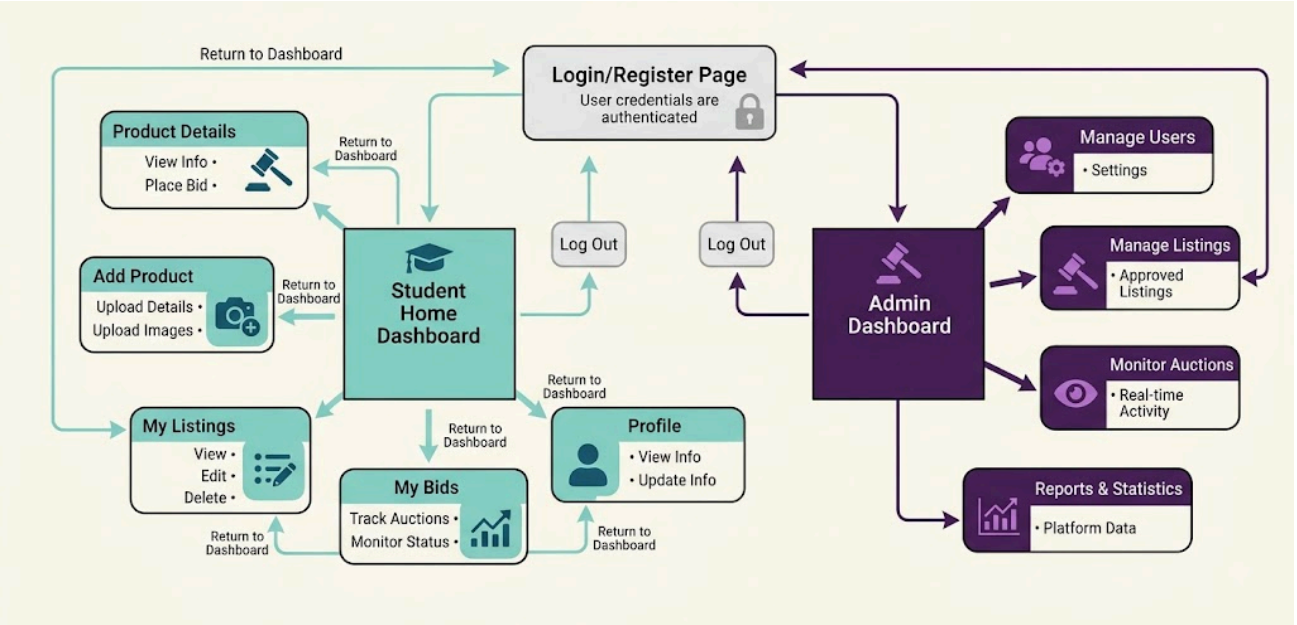
- Manage Users
- Manage Listings
- Remove Inappropriate Content
- View Reports and Statistics

5. Views / Pages

View Name	Purpose	Components	Navigation	Expected Output
Login Page	User authentication	Login form	Login → Dashboard	User successfully logs in
Register Page	Create new account	Registration form	Register → Login	New account created
Home/Dashboard	Display ongoing auctions	Product cards, search bar, filters	Navigate to Product Details, Profile	List of active auctions
Product Details	View product information	Images, description, current bid, bid form	Bid, Back to Dashboard	Detailed product information
Add Product	Create a new auction listing	Product form, image upload	Submit → Dashboard	Product listed successfully
My Listings	View/manage own products	Product list, Edit/Delete buttons	Edit/Delete Product	Updated product list
My Bids	View auctions user has bid on	Bid history	Product Details	List of bids placed
Profile Page	Manage account	User details, edit profile	Dashboard	Updated profile

Admin Dashboard	Manage platform	User list, product list, reports	Manage Users/Listings	Updated platform records

6. Navigation Flow



The application begins with the **Login/Register** page, where students can either create a new account or log in using their credentials. After successful authentication, users are redirected to the **Home Dashboard**, which displays all active auction listings.

From the **Home Dashboard**, users can:

- Open the **Product Details** page to view complete information about a product and place a bid.
- Navigate to the **Add Product** page to create a new auction listing by uploading product details and images.
- Access the **My Listings** page to view, edit, or delete their own auction listings.
- Open the **My Bids** page to track auctions they have participated in and monitor their bid status.
- Visit the **Profile** page to view and update their personal information.
- Log out of the application.

Administrators log in through the same authentication page but are redirected to the **Admin Dashboard** after successful login. From there, they can:

- Manage user accounts.
- Approve, edit, or remove product listings.
- Monitor ongoing auctions.

- View reports and platform statistics.
- Log out when finished.

Overall, the **Home Dashboard** acts as the central hub of the application, allowing users to easily navigate between auction-related features and return to the main page after completing any action.

7. Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript, TailWind, React
Backend	Node.js with Express.js
Database	MongoDB, Cloudflare R2
IDE	Visual Studio Code
Version Control	Git & GitHub